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# Practical 1: Introduction to Deep Learning Frameworks
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import tensorflow as tf
from tensorflow.keras import layers, models
import numpy as np
import matplotlib.pyplot as plt

# 1. Framework Verification & Tensor Exploration
print("TensorFlow Version:", tf.__version__)

# Create and inspect basic Tensors
scalar_tensor = tf.constant(5)
matrix_tensor = tf.constant([[1.0, 2.0], [3.0, 4.0]], dtype=tf.float32)

print("\n--- Tensor Operations ---")
print("Scalar Tensor:", scalar_tensor)
print("Matrix Tensor Shape:", matrix_tensor.shape)
print("Matrix Multiplication Result:\n", tf.matmul(matrix_tensor, matrix_tensor).numpy())

# 2. Synthetic Data Preparation (Linear Relationship:  $y = 2x - 1$ )
X_train = np.array([-1.0, 0.0, 1.0, 2.0, 3.0, 4.0], dtype=float)
y_train = np.array([-3.0, -1.0, 1.0, 3.0, 5.0, 7.0], dtype=float)

# 3. Create Basic Neural Network Architecture
# Demonstrating layers (Dense) and activation functions
model = models.Sequential([
    layers.Input(shape=(1,)),
    layers.Dense(units=16, activation='relu', name="hidden_layer"),
    layers.Dense(units=1, activation='linear', name="output_layer")
])

# 4. Model Compilation
# Demonstrating Optimizers (Adam, SGD) and Loss Functions (MSE)
model.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=0.01),
    loss='mean_squared_error',
    metrics=['mae']
)

model.summary()

# 5. Model Training
print("\n--- Training Model ---")
history = model.fit(
    X_train,
    y_train,
    epochs=250,
    verbose=0
)
print("Training Complete!")

# 6. Evaluation and Inference
test_input = np.array([10.0])
predicted_output = model.predict(test_input)
print(f"\nInference: For input x = 10.0, Expected y ~ 19.0, Predicted y = {predicted_output[0]}")

# 7. Visualize Training Loss Curve
plt.figure(figsize=(8, 4))
plt.plot(history.history['loss'], label='Training Loss (MSE)')
plt.title('Loss Reduction Over Epochs')

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plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.grid(True)
plt.legend()
plt.show()
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TensorFlow Version: 2.21.0

--- Tensor Operations ---
Scalar Tensor: tf.Tensor(5, shape=(), dtype=int32)
Matrix Tensor Shape: (2, 2)
Matrix Multiplication Result:
[[7. 10.]
 [15. 22.]]
Model: "sequential"

Layer (type)	Output Shape	Param #
hidden_layer (Dense)	(None, 16)	32
output_layer (Dense)	(None, 1)	17

Total params: 49 (196.00 B)
Trainable params: 49 (196.00 B)
Non-trainable params: 0 (0.00 B)

--- Training Model ---
Training Complete!
1/1 ————— 0s 63ms/step

Inference: For input x = 10.0, Expected y ~ 19.0, Predicted y = 16.3393



